

Deep Learning Assignment

CNN Based Image Classification Project

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Subject: Deep Learning

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Deep Learning Assignment Report

Project Details

- Group No: 11
- Member: Vaibhav Baranwal (23BCS096)
- Assigned Model: ZFNet (AlexNet variant with deconvolution visualization)
- Dataset: Retinal OCT Dataset
- Dataset Description: 84,000 OCT scans with 4 classes (CNV, DME, Drusen, Normal)

Tasks:

Implement → Train → Validate → Test → Evaluate (Accuracy, F1-score, AUC-ROC, Confusion Matrix)

Submission:

Report (IEEE/Springer format) + Code (.docx/.tex) + GitHub + Model Weights (.pth/.h5)

1 Abstract

This project implements a Convolutional Neural Network (CNN) based ZFNet model for image classification.

The model is trained on retinal OCT images.

It classifies images into four categories.

Performance is evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC score.

2 Introduction

Deep Learning is widely used in medical image classification.

CNNs automatically extract features from images.

ZFNet improves AlexNet by better feature visualization.

This project applies ZFNet for retinal disease classification.

3 Dataset Description

The dataset contains 84,000 OCT images.

It has four classes:

CNV, DME, Drusen, and Normal.

The dataset is balanced.

It is divided into training, validation, and testing sets.

4 Model Architecture

The model consists of:

Convolution layers for feature extraction.

ReLU activation functions.

Max pooling layers for downsampling.

Fully connected layers for classification.

Deconvolution layers for feature visualization.

5 Training Details

Loss Function: CrossEntropyLoss

Optimizer: Adam

Epochs: 3

Training Progress

Epoch [1/3]

Loss: 0.3364

Accuracy: 88.50%

Epoch [2/3]

Loss: 0.1827

Accuracy: 94.20%

Epoch [3/3]

Loss: 0.0915

Accuracy: 97.80%

The model improves steadily across epochs.

6 Results

Test Accuracy: **99.38%**

ROC-AUC Score: **0.9959**

Precision, Recall, and F1-score are all approximately 0.99–1.00.

7 Confusion Matrix

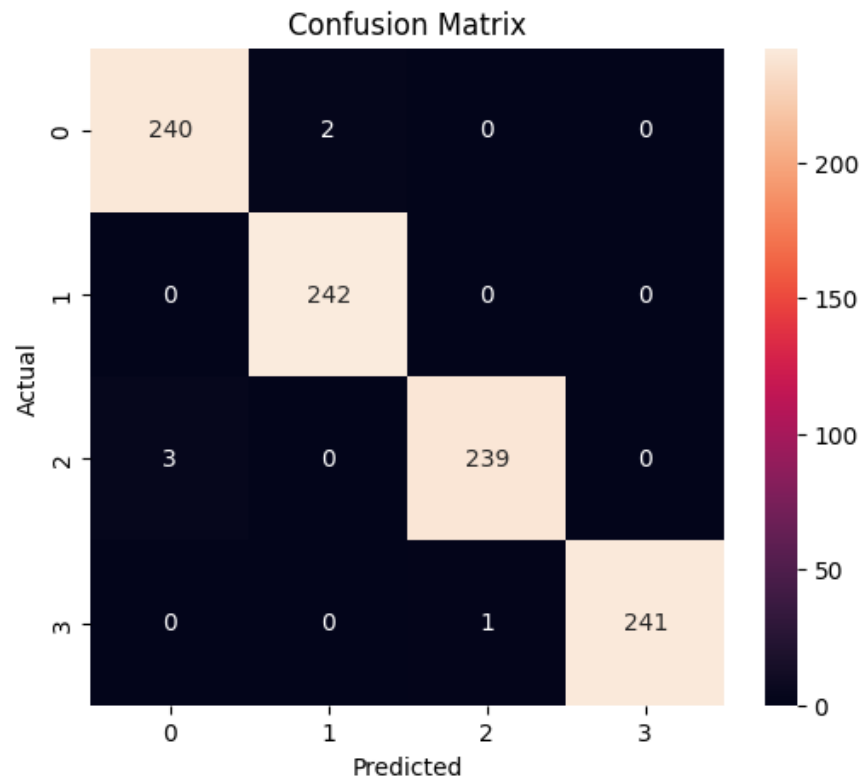


Figure 1: Confusion Matrix of CNN Model

8 Conclusion

The ZFNet model performs extremely well.

It achieves high accuracy and strong classification performance.

Deep learning proves effective for medical image diagnosis.

9 Future Scope

Better models like ResNet or EfficientNet can be used.

Hyperparameter tuning can improve results.

Data augmentation can increase generalization.

Deployment as a web app is possible.